

An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

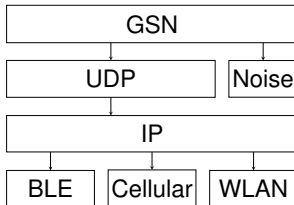
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Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

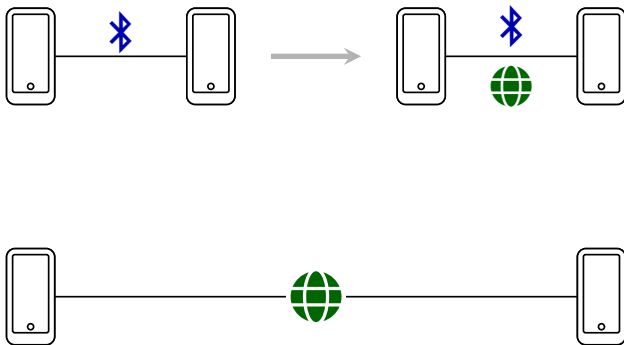


Architecture



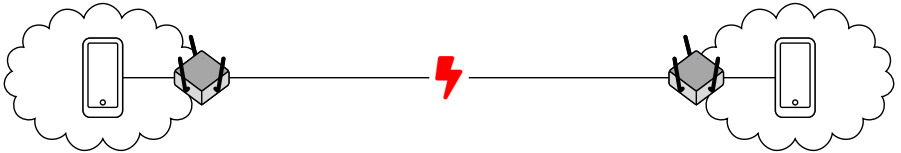
- Connect two peer devices according to priority and proximity pair
- Dual-stack network comms: BLE first, Cellular (5G/LTE) and WLAN second
- In close range, use BLE for signaling to establish UDP connection
- ICE-similar stack for address reflection, hole-punching using friends

Dual-stack comms

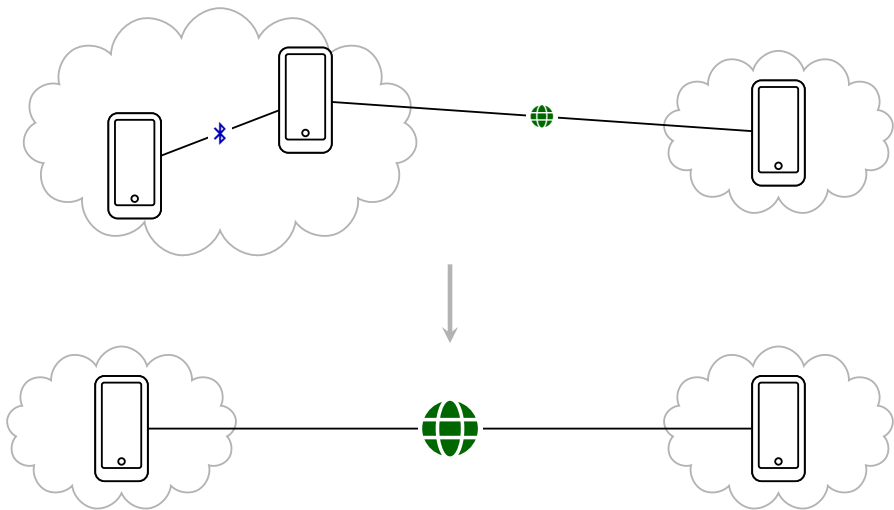


- BLE as signaling channel
- Advertise multiple addresses
- UDP for hole-punching
- UDX for long-range communication

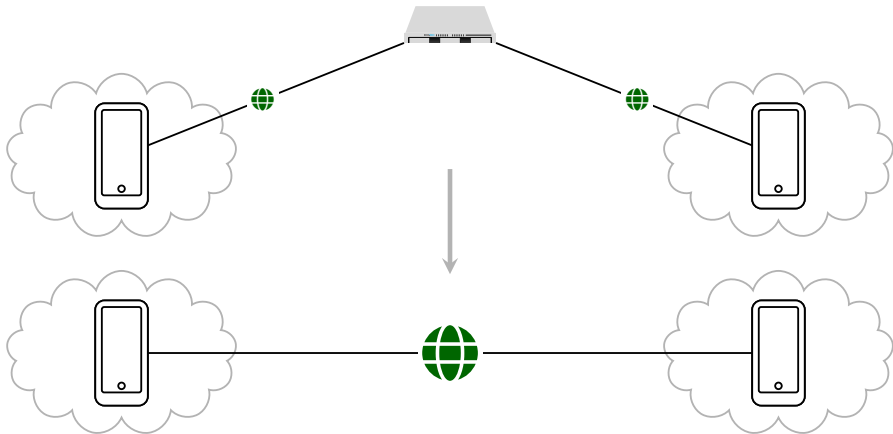
NAT



Well-connected friends



Rendezvous Server



Since mid-April: Flutter project work

- Migrated BLE from separate central/peripheral plugins to one `grassroots_bluetooth_layer`: unified event model, shared discovery UUID, post-connect ANNOUNCE identity, and BLE role toggle.
- Reworked UDX into a dual-stack Internet transport: IPv4/IPv6 sockets, multi-candidate addresses, live network/public-address refresh, and best candidate-pair selection.
- Aligned rendezvous/NAT traversal with GLP: RECONNECT/AVAILABLE matching, two-sided PUNCH_INITIATE, `addrReflect` registration proof, and RV fan-out.
- Added well-connected-friend and friends-of-friends mediation: friends can facilitate rendezvous only when the target was advertised through the friend graph.
- Moved friend-shared rendezvous lists from `PeerState` to `FriendshipState` so `RV_LIST` data is friendship-scoped, persisted, and survives restart.
- Secured signaling with Noise XX across clients and rendezvous anchors, removing the plaintext configured-RV path.
- Hardened delivery over UDX/BLE: ACKs after fragment reassembly, messageId through fragments, UDX packet framing/u32 lengths for photos, and stale-session/socket recovery.

Mobility model: city-scale BLE relay

Goal

- Test if BLE relay can replace UDP locally.
- Compare direct, friend, and open relay (simulated).
- Report delivery, latency, overhead, and abuse surface.

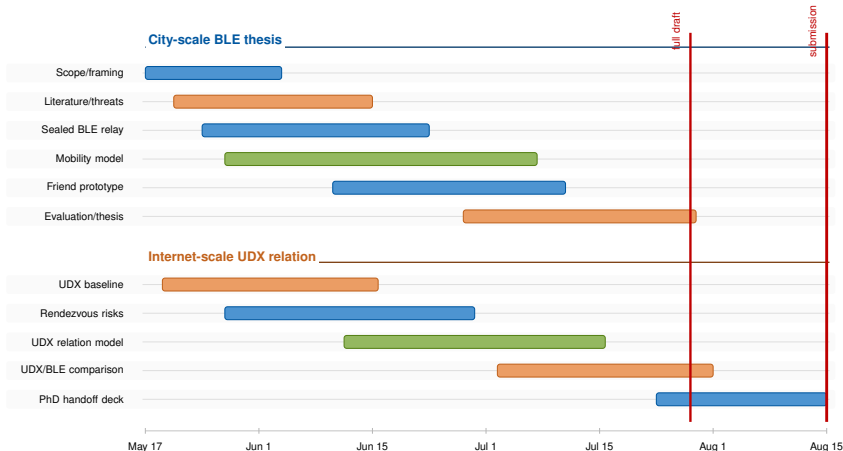
What we track

- Encounters: pseudonym pair, start/end, duration, inter-contact, RSSI band.
- Delivery: created, relayed, ACKed, hop count, TTL expiry, size, failure.
- Cost: relay bytes, duplicates, queues, scan/advertise time, battery proxy.

How

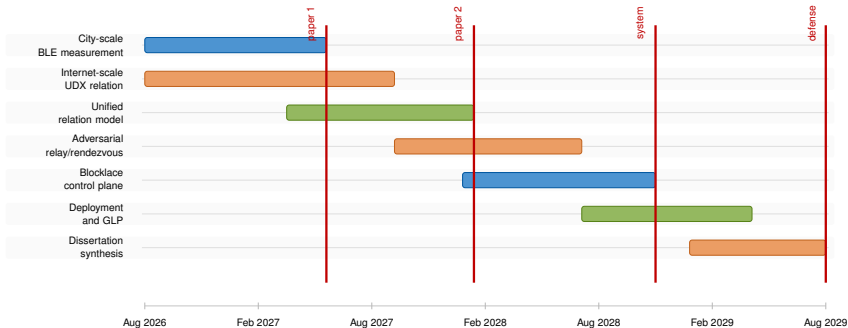
- Encrypted local trace log.
- Upload only by explicit opt-in.
- Replay traces in simulator calibrated by BLE contacts.
- No content; coarse location only by separate opt-in.

Master gantt: thesis plus UDX track



- Thesis core: BLE mobility and secure local relay; Keidar–Shapiro thread: keep UDX as an Internet-scale relation model and comparison.

PhD runway: two relation systems



- Thesis-to-PhD bridge: compare what grassroots relations mean under local physical mobility and under Internet-scale UDX connectivity.